

Signals and Systems Operations

Participated in & did the coursework [Y/N]?

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1 INTRODUCTION

This laboratory exercise focuses on the fundamental operations of continuous time signals and systems. Specifically, the objective of the exercise is to mathematically define, simulate, and visualize operations of continuous time signals and systems such as cross-correlation and convolution using computational tools.

In order to attain the objective signals are represented mathematically using symbolic expressions. The Heaviside step function is useful in this context as it allows for the precise definition of piecewise continuous signals by acting as a mathematical switch [1]. To mathematically measure the similarity between these signals, cross-correlation is used. It measures the similarity between two distinct signals as one is shifted relative to the other. By evaluating the different cases of these overlapping intervals the features of a reference signal that aligns with the other signal over time is determined [2]. Meanwhile, convolution is used to determine how a system reacts to an input signal based on its impulse response. The resulting output response, $y(t) = x(t) * h(t)$ provides a complete behavioral model of a linear time-invariant system [2].

Utilizing matlab to apply these concepts, several specific MATLAB functions are needed. First, the syms function which initializes symbolic variables. Second, heaviside() which constructs the discrete transitions in the signal amplitudes. Third, the int() function which computes the area under the overlapping signals. Fourth, subs() which replaces symbolic variables to evaluate specific time shifts. Moreover, visualization functions are needed to graphically interpret the results where functions such as fplot() to graph the symbolic expressions over specified domains, subplot() to display the different plots side-by-side, title() to add a descriptive title to the plots, and grid on to organize multiple graphs into clear, readable figures are used.

2 EXPERIMENTAL DESIGN

The experimental approach for this study centers on the computational modeling and verification of fundamental signal operations—specifically cross-correlation and convolution—using the

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MATLAB environment. By transitioning from symbolic mathematical definitions to graphical system responses, the design allows for an analytical comparison of how different signals interact over time. The methodology is structured into three primary phases: signal definition, operational segmentation, and computational execution.

2.1 Mathematical Signal Definitions

The initial stage of the experiment involves the precise mathematical construction of the signals using the MATLAB Symbolic Math Toolbox. Signals are constructed using the `heaviside(t)` function to define specific temporal windows. For the cross-correlation phase, the “Target Signal” $x(t)$ is established as a ramp function bounded between $t = -2$ and $t = 2$, defined by $x(t) = t \cdot [u(t+2) - u(t-2)]$. The “Search Signal” $y(t)$ is a reflected and scaled ramp defined as $y(t) = -2t \cdot [u(t+1) - u(t-1)]$.

For the convolution phase, which simulates a system response, a pulse signal $x_{in}(t)$ is created using the expression $x_{in}(t) = u(t) - 2u(t-1) + u(t-2)$. This input is paired with a first-order system impulse response, $h(t) = e^{-10t}u(t)$, representing a rapidly decaying exponential characteristic.

2.2 Segmentation of Interaction Boundary Conditions

To calculate the total response, the experimenter must identify the critical time-shifts (τ) where the overlapping area between the signals changes its geometric behavior. The following table summarizes these segments to guide the symbolic integration process:

Case	Interaction Phase	Interval Boundary	Integration Limits
1	Pre-Interaction	$\tau < -3$	Integral = 0
2	Signal Entry	$-3 \leq \tau < -1$	2 to $\tau + 1$
3	Full Immersion	$-1 \leq \tau \leq 1$	$\tau - 1$ to $\tau + 1$
4	Signal Exit	$1 < \tau \leq 3$	$\tau - 1$ to 2
5	Post-Interaction	$\tau > 3$	Integral = 0

Table 1

Boundary conditions for cross-correlation integration.

2.3 Step-by-Step Computational Procedure

The following logic flow serves as a procedural roadmap for executing the experiment in a MATLAB environment:

- 1) Initialize Variables:** Define the independent and shifting time variables using `syms t tau real`.

- 2) **Define Core Functions:** Code the mathematical models for $x(t)$, $y(t)$, and $h(t)$ using Heaviside functions.
- 3) **Calculate Cross-Correlation:** Implement the integral $R_{xy}(\tau) = \int x(t)y(t - \tau)dt$ using the `int` function for each of the five boundary cases.
- 4) **Synthesize Results:** Consolidate the individual case integrals into a single unified function using the `piecewise` command.
- 5) **Simulate System Output:** Execute the convolution integral $y_{out}(t) = \int x_{in}(\tau)h(t - \tau)d\tau$ for $t \geq 0$.
- 6) **Visualize Data:** Generate high-resolution plots using `fplot` with appropriate axis labels and titles.

2.4 Procedural Flowchart and Algorithmic Logic

The experimental design is visually summarized by the following logical progression:

Start → **Define Symbolic Signals** → **Boundary Identification**
 → **Symbolic Integration** → **Piecewise Reconstruction** →
Graphical Rendering → **End**

The underlying algorithm for the convolution output response $y(t)$ is implemented as follows:

```
IF t < 0:
    Output = 0
ELSE IF t >= 0:
    Output = Integral of (x(tau) * exp(-10*(t-tau))) from 0 to t
Render Plot on Interval [-5, 5]
```

3 ANALYSIS OF RESULTS

The results obtained from the lab activity provide a comprehensive look at the mathematical manipulation of signals and the resulting waveforms. This section consolidates the findings to demonstrate the practical application of signal operations and system modeling in a computational environment.

3.1 Time-Domain Transformations of the Baseline Signal

The first phase of the analysis focused on the manipulation of the baseline signal $x(t)$, which is a linear ramp function defined within the interval $t \in [0, 8]$. As observed in the consolidated results, the original signal transitions from an amplitude of 1 at $t = 0$ to an amplitude of -1 at $t = 8$. The transformation $x(t - 2)$ demonstrated a successful time delay, where the entire waveform shifted two units to the right without altering its shape or amplitude. Conversely, $x(t + 4)$ showed a time advance, moving the signal four units to the left into the negative time domain. The time-reversal operation $x(-t)$ resulted in a vertical reflection across the y-axis, effectively flipping the ramp's orientation. The combined operation $x(3 - t)$ proved to be the most complex, as it required both a reversal and a 3-second shift, resulting in a signal that peaks at $t = 3$. Finally, the time-scaling operation $x(2t)$ compressed the original 8-second duration into a 4-second window, confirming that a scaling factor greater than unity increases the "speed" of the signal relative to the time axis.

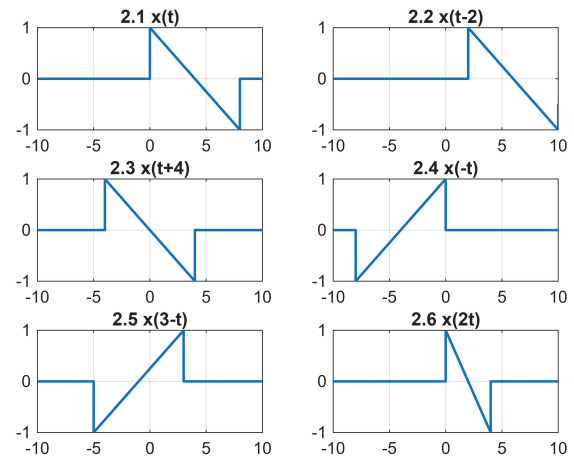


Figure 1. Subplots of Signal Transformations illustrating shifting, reversal, and scaling.

3.2 Piecewise Evaluation of Composite Signal $y(t)$

The composite signal $y(t) = e^{-t}u(t) + u(-1 - t)$ was analyzed to observe the interaction between decaying and constant signal components. The resulting plot reveals three distinct operational regions that align with theoretical expectations. For $t < -1$, the signal maintains a constant unit amplitude of 1, governed by the time-reversed and shifted unit step function $u(-1 - t)$. A critical observation is the "dead zone" or gap between $t = -1$ and $t = 0$, where neither part of the piecewise function is active, resulting in an amplitude of exactly zero. At $t = 0$, the signal abruptly returns to an amplitude of 1 and immediately begins a smooth exponential decay toward zero, as dictated by the $e^{-t}u(t)$ term. This simulation confirms the system's ability to model signals with high-frequency discontinuities and differing temporal characteristics in a single expression.

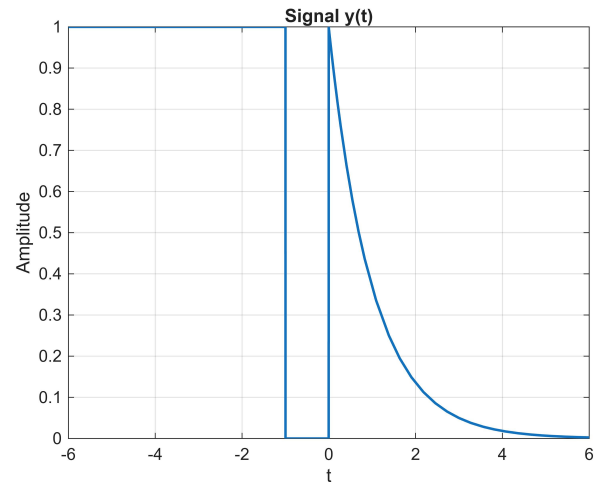


Figure 2. Visualization of composite signal $y(t)$ highlighting piecewise discontinuities.

3.3 Complementary Impulse Responses of Filter Systems

The comparative analysis of the impulse responses for the lowpass and highpass filters highlights the mathematical relationship between different system types. The lowpass response

$h_1(t)$ is characterized by a positive exponential decay that starts at 1 and approaches zero as time increases, which effectively models the attenuation of high-frequency components. In contrast, the highpass response $h_2(t)$ —when plotted excluding the initial Dirac delta impulse—starts at a negative amplitude of -1 and decays upward toward the zero-axis. This inverse relationship is expected, as the highpass filter in this experiment is mathematically derived by subtracting the lowpass response from the input impulse. The visualization confirms that both filters reach a steady state of zero within approximately 0.6 seconds, indicating a stable system response.

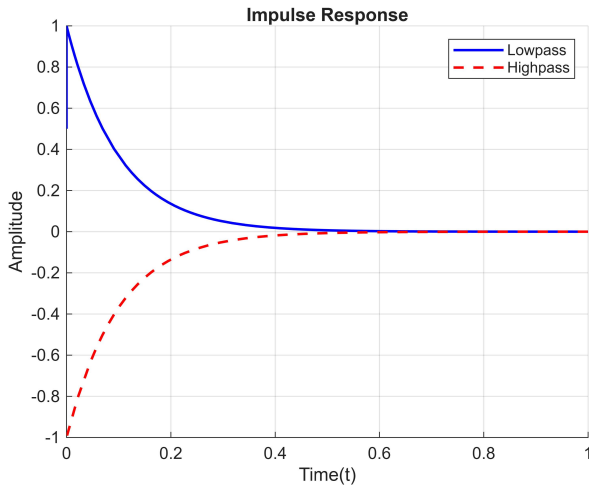


Figure 3. Overlay of filter impulse responses demonstrating complementary decay behaviors.

3.4 Statistical Evaluation of Auto-correlation $R_{xx}(\tau)$

The auto-correlation plot provides a statistical measure of how much the signal $x(t)$ overlaps with itself at various time lags. The resulting waveform is perfectly symmetric around the vertical axis $\tau = 0$, where the peak amplitude of approximately 2.7 occurs. This peak represents the total energy of the signal. As the lag τ increases or decreases, the correlation values transition from positive to negative, specifically forming two distinct “lobes” that reach a minimum of approximately -1.1. These negative values occur because the ramp signal $x(t)$ contains both positive and negative amplitudes; when shifted by certain lags, the positive portions of the original signal align with the negative portions of the shifted version, resulting in a negative correlation product. The correlation eventually returns to zero when the lag exceeds ± 8 seconds, as there is no longer any overlap between the two versions of the signal.

3.5 Numerical Results and Dataset Characteristics

To ensure the accuracy of the simulations, specific numerical parameters were maintained across all group members’ experiments. The transition from symbolic expressions to numerical data utilized a sampling interval of $dt = 0.01$ seconds, which equates to a sampling frequency of 100 Hz. This resolution was sufficient to capture the rapid transitions in the filter impulse responses and the slope of the ramp function without significant aliasing. The numerical array `x_num` was generated over a 20-second window, resulting in a vector of 2001 data points. Consequently, the auto-correlation result `Rxx` produced a vector of 4001 points, adhering to the standard signal processing rule

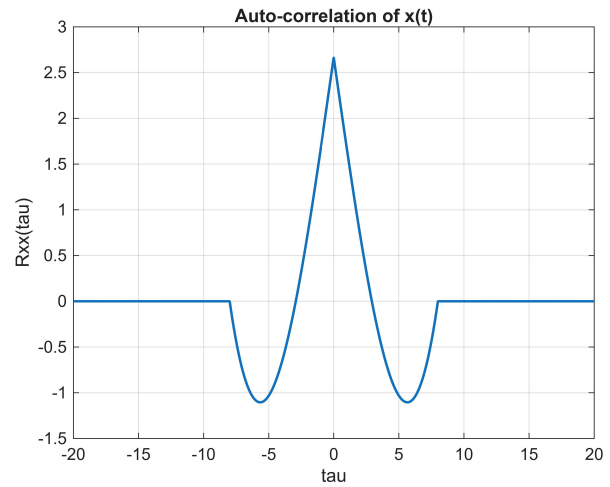


Figure 4. Auto-correlation function highlighting temporal symmetry and peak energy at zero lag.

where the length of a correlation or convolution is $2N - 1$. The final amplitude of the auto-correlation peak (~ 2.7) was achieved by multiplying the raw `xcorr` output by the time step dt , which correctly approximates the continuous-time integral of the signal's square.

Table 2
Consolidated Numerical Parameters and Simulation Results

Parameter	Group Consensus Value	Description
Sampling Interval (dt)	0.01	Time resolution for numerical arrays
Time Range	-10 to 10 seconds	Total window for signal observation
Input Vector Size	2001 samples	Dimensions of <code>x_num</code>
Correlation Vector Size	4001 samples	Dimensions of <code>Rxx</code>
Max Correlation ($R_{xx}(0)$)	~ 2.7	Peak energy value at zero lag

4 DISCUSSION AND CONCLUSION

The laboratory exercise successfully demonstrated the mathematical modeling, manipulation, and characterization of continuous-time signals and systems using MATLAB, fulfilling the primary objective to define and visualize these sequences. The first phase of the experiment confirmed that temporal manipulations, specifically time shifting, flipping, and scaling, directly alter a signal's independent variable to achieve predictable geometric changes. As evidenced in the results, the baseline signal $x(t)$, which originally spanned from $t = 0$ to $t = 8$ seconds, was successfully shifted, reflected, and compressed; for instance, the operation $x(2t)$ halved the signal duration to 4 seconds. These transformations validated the foundational theory of Linear Time-Invariant (LTI) systems by showing how signal geometry is modified through precise mathematical operations [1].

Furthermore, the characterization of systems through impulse responses $h(t)$ provided critical insight into filter behavior and frequency selectivity. By implementing symbolic equations for lowpass, highpass, bandpass, and bandstop filters, the experiment showcased how different systems react to an instantaneous unit impulse. A direct comparison between the lowpass and highpass responses revealed a notable symmetry: the lowpass response $h_1(t)$ exhibited a positive exponential decay starting at an amplitude of 1, while the highpass response $h_2(t)$, excluding the initial impulse, mirrored this with a negative decay starting at

–1. This practical application of the Dirac delta $\delta(t)$ and Heaviside step $u(t)$ functions in MATLAB reinforced the theoretical understanding of how LTI systems are entirely defined by their impulse responses [2].

Finally, the experiment explored internal signal analysis through the computation of auto-correlation $R_{xx}(\tau)$, which measured the signal's temporal similarity to itself. By transitioning from symbolic expressions to numerical data arrays with a high-resolution time step of $\Delta t = 0.01$, the total energy of the signal was identified at the zero-lag peak, reaching an amplitude of approximately 2.7. The resulting symmetric plot confirmed that as the time lag τ increases, the correlation drops, eventually returning to zero when the waveforms no longer overlap. In conclusion, the integration of MATLAB's computational tools effectively bridged theoretical concepts with practical simulation, providing a robust framework for analyzing the interaction between signals and systems [3].

5 AUTHORS CONTRIBUTION

On what contributions to specify, see the terms at www.elsevier.com.

- 1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration.
- 2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision.
- 3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing.

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